

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

☎ (587) 879-4626 | ✉ tkuipers123@gmail.com | 🏠 tkuipers.ca | 📱 tkuipers | 🌐 tyler-kuipers

About Me

I've spent 10 years building backend systems, with the last 3 focused on distributed systems architecture and cloud infrastructure. I design multi-account AWS architectures, run production Kubernetes clusters, and build the tooling that makes teams faster. I built a test framework that's been used by our team for 4+ years and reduced defects by 30%. I've worked in Python, Java, and TypeScript, and I'm looking to deepen my skills in Go. I'm drawn to high-scale systems and the reliability challenges that come with them.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Architected modernization of document generation system from legacy C# to containerized microservices on EKS. Integrated Datadog for metrics, logging, and alerting. Led 2-3 engineers. Reduced environment provisioning from weeks to minutes. (Kubernetes, Docker, Datadog, Helm, CI/CD)
- Architected technical infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts with external partner integration. 10 Lambda functions, multi-million dollar annual revenue, 2.8M events processed, 99.9% uptime. Won 2024 Guidewire Innovation Award. (AWS, Terraform, Lambda, Python)
- Built self-service infrastructure platform that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Includes database snapshotting for production issue diagnosis. Built solo, adopted by 15 engineers across 3 teams. (Python, Terraform, AWS, Jenkins, Docker)
- Own API architecture for platform serving 5 internal teams and external partners. Approve endpoint additions, maintain design standards, and ensure consistency across services. (Python, REST, OpenAPI)
- Set coding standards and maintain technical documentation for 12-person team. Author RFCs for cross-team initiatives. Member of Architecture Steering Committee reviewing proposals org-wide.
- Mentor engineers on debugging complex distributed systems issues. Trained 5-6 developers on TDD methodologies.

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built backend services running medical AI algorithms, optimizing for low latency and high throughput. (TypeScript, Node.js, AWS)
- Improved algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, identifying and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Built automated test framework that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later. (Java/Gosu, JUnit)
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local dev environments.
- Led 5-6 person development team using TDD methodology. Delivered projects on time with no critical defects post-launch.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

SOFTWARE ENGINEER ASSOCIATE

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

Edmonton, Alberta

May 2014 - Sep 2016

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.

Education

The Open University
MSC. COMPUTING (SOFTWARE ENGINEERING)
Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)
May 2024 - Ongoing

The King's University
B.S. IN COMPUTER SCIENCE

Edmonton, Alberta
Sep 2012 - Dec 2016

Skills

System Design	Design distributed systems with multi-account AWS architectures, Kubernetes, and Terraform. Built systems serving 5 teams that process \$4.3M in annual transactions with 99.9% uptime.
Technical Leadership	Own architectural decisions for 12-person engineering team. Approve API endpoint additions, write RFCs, and drive technical decisions that multiple teams follow.
Cross-Team Coordination	Coordinate across internal teams and external partners on multi-team initiatives. Designed systems used by 5 internal teams and external partners.
Platform Engineering	Built self-service infrastructure platform (60x faster provisioning) and test frameworks that reduced defects 30%. Focus on multiplying team effectiveness.

Technologies

Languages	Python, TypeScript, Java, Bash
Backend	REST APIs, OpenAPI, Microservices, CI/CD
Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Kubernetes, Docker, Terraform, Helm
Observability	Datadog, Elasticsearch, Kibana
Databases	PostgreSQL, Aurora/RDS, Redis